

Transferring a Dex3-1 manipulation policy to BrainCo Revo2 Touch hands

A frozen GR00T N1.7 policy trained on a Unitree G1 29-DoF with 7-DoF Dex3-1 hands, made to drive the same robot wearing 6-DoF BrainCo Revo2 Touch hands, without retraining and without changing the policy's action space. Each step folder has its own README with the commands; this is what was done and what came out of it.

Platform and task

The requirement was a pretrained policy whose hands could be swapped for a target with a *different* DoF count, so that retargeting is a real problem rather than a relabelling. A GR00T N1.5 GR1 tabletop task ran but is a 6→6 swap on the wrong robot, and RoboCasa has no G1 with a dexterous-hand checkpoint. The G1 locomanipulation apple task in MuJoCo was abandoned after an A/B test traced its failures to the locomotion controller rather than the policy, and a fixed-base variant built to remove locomotion still scored 0%; with no working reference there is nothing to retarget from.

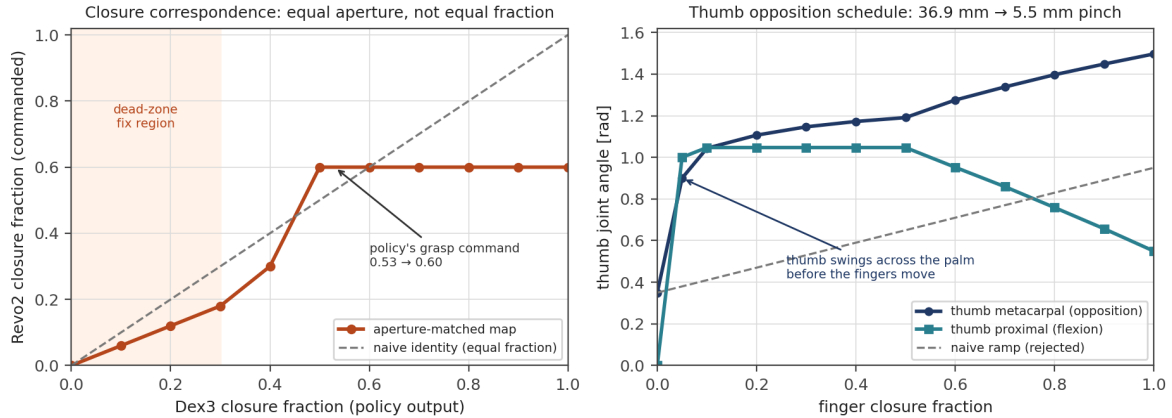
The choice was IsaacLab-Arena's `galileo_g1_static_pick_and_place`: correct robot, 7 DoF per hand against the Revo2's 6, no locomotion, a published checkpoint, and a baseline reproducing at 0.65. Two limits are worth stating. The task is single-handed in practice — the right-hand action is identically zero, so although the retargeting layer is symmetric and both hands are tested, only the left runs in closed loop. And the apple spawn is deterministic, so this measures repeatability, not spatial generalisation.

What the two hands differ by

The policy runs at 50 Hz on a 480×640 head camera, a 43-D joint state and a 50-D action, feeding an AGILE whole-body controller. None of that changes between the two runs; the policy is not aware the hands changed.

The Dex3-1 has three fingers and seven independent actuators. The Revo2 has five fingers and six, with five distal joints coupled through URDF `<mimic>` tags. Beyond the count: fingertip positions at equal closure fraction differ by 2–3 cm; ring and pinky have no source signal at all and `thumb_0` has no target counterpart; the Revo2 takes about 0.65 s to close fully against the Dex3's ~0.2 s; and authored hand mass drops from 1.39 kg to 0.29 kg. Maximum fingertip aperture is 193.3 mm against 140.6 mm, and the grasp centres sit 25.2 mm apart. Those numbers come from forward kinematics on both URDFs rather than datasheets, and `2_retarting/analyze_hand_geometry.py` reproduces them.

The correspondence



Angle conventions are irreconcilable — Dex3 left and right joints are sign-mirrored and several have negative "closed" limits, while every Revo2 joint runs 0 to positive — so the map is defined on *closure fraction*. Index, middle and thumb collapse to one grasp closure, since the Dex3 pinches as a unit here. Ring and pinky follow the mean of index and middle, the one place the map invents information not in the source.

Four things then have to be right at once, each found through a failure:

- **Aperture calibration.** Equal fraction is not equal opening: matching fractions leaves a 33 mm gap where the Dex3 pinches to 11.7 mm, so the hand closes on empty air. The table matches openings on the *running minimum*, because both aperture curves turn back up once the fingers curl past the thumb, and matching the instantaneous value would let squeezing harder re-open the hand.
- **Thumb opposition.** The Revo2's metacarpal sweeps across the palm rather than rolling, so a searched schedule puts it into opposition ahead of the fingers. Pinch aperture at 75% closure goes from 36.9 mm to 5.5 mm; without it no grasp is geometrically possible.
- **Invertibility.** Strict matching wants zero Revo2 closure below Dex3 closure ≈ 0.35 , and that flat segment is a closed-loop hazard: the policy commands a closure, nothing moves, and its next observation reports an open hand. It showed in logs as the policy opening and re-closing mid-grasp. Lifting those entries took round-trip error from 0.300 to 0.000.
- **Interface preservation.** The action space stays 50-D in Dex3 coordinates and the 41-DoF articulation is reached only inside the action term, so the policy, its client, the modality config and the controller are untouched.

The wrist adapter belongs here too. The policy aims the *wrist*, having learned where a Dex3 palm sits relative to it, and at the stock mount the Revo2's grasp centre is 25.2 mm away, so the hand closes above the apple's equator and rolls it. That offset is applied in the URDF; on hardware it is a machined plate. Orientation was left alone — grasp quality proved invariant to mount rotation.

The asset

The URDF got two edits: the wrist adapter, and a restored `pelvis_contour_link` that the supplied file omits. Isaac Lab's converter is then unfaithful in three ways, all fixed: it welds the pelvis to the world despite `fix_base=False`, so the robot never stands; it marks every link instanceable, and USD traversal does not descend into instanced prims, so the task's high-friction material silently never binds to the fingers; and it authors no `maxJointVelocity`, leaving the body joints effectively unlimited where the stock G1 carries caps.

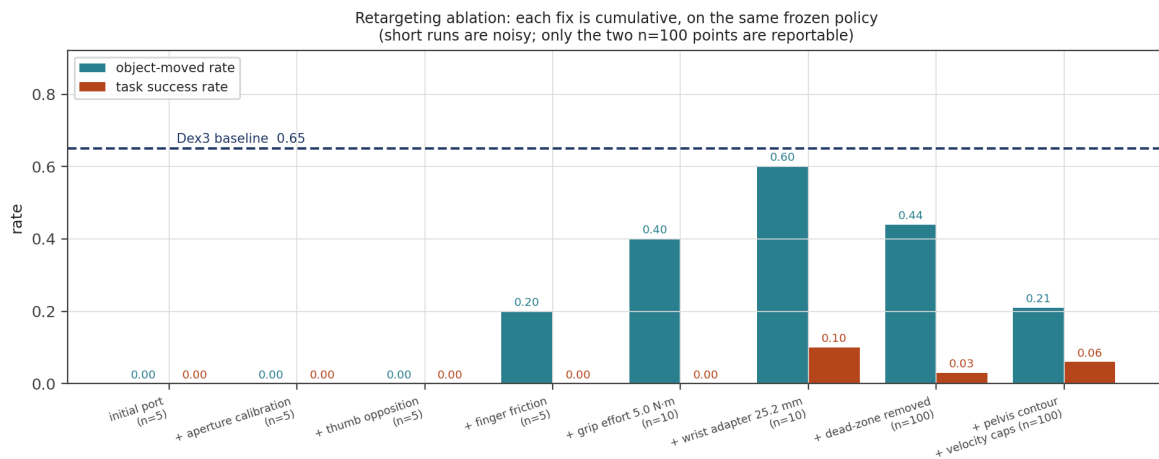
A fourth difference produced no error at all and would have invalidated the comparison. The stock G1 is authored flat, so the default camera path `Robot/head_link/RobotHeadCam` resolves; the converter nests links under a `Geometry` scope, that path matches nothing, and USD quietly creates an empty `Xform` at the robot origin instead of failing. The head camera sat at the pelvis staring at the robot's own arms — and the policy is a vision model. Fixed, then verified against the baseline stream.

Finger effort was set to 5.0 N·m to match the baseline hand group rather than the URDF's 0.5–2.0, which is the outlier: BrainCo rate the hand at 50 N grasp. Velocity limits stay at datasheet values, so the slower closing survives as a genuine difference.

Results

	Dex3-1 baseline	Revo2, retargeting only
Task success rate	0.65 (13/20)	0.06 (6/100)
Frame-verified pick-and-place	not audited	0.06, all six genuine
Object moved rate	0.65	0.21

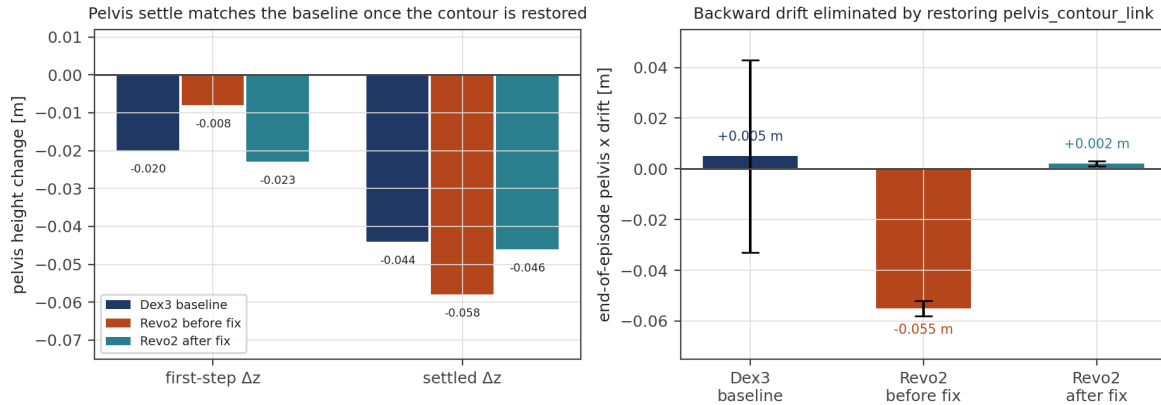
Identical policy, task, scene, success criterion and evaluation script; the only difference is `--embodiment`.



Only the two 100-episode rows are reportable — one configuration scored 1/10 and then 0/20 on repeat. Object-moved is the more informative column: it separates "the hand never reaches the apple" from "it reaches but cannot hold it", and the first three configurations touch the apple in zero episodes. The honest headline is that **no gripper-side change alone produced a single success**. The two that moved the needle were the wrist adapter, which is mount geometry, and the pelvis collider, which is not a hand change at all.

Success also has to be verified by eye. The task counts the apple being *pushed* into the plate region; in one 100-episode run only 1 of 3 counted successes was a real grasp, so every number above was checked frame by frame at 4 fps.

The pelvis



The most instructive failure. The pelvis crept 5.7 cm backwards every episode and settled 2.4 cm lower, where the baseline settles within 0.4 s around a centred +0.5 cm mean — which turns the top-down grasp the policy learned into a side sweep that pushes the apple instead of lifting it.

Everything authored matched: gains, armature, friction, limits, every link mass. Comparing the runtime values PhysX actually loaded found it — the scene has an invisible collision slab under the table, the robot spawns overlapping it, and the stock G1 braces against it with a pelvis_contour_link collider that the supplied BrainCo URDF omits entirely. What looked like drift was the free-standing controller doing its job. Restoring the link took true pick-and-place from 1/100 to 6/100.

Getting there meant rejecting more plausible explanations, which is where most of the time went: inherited baseline behaviour (the baseline does not drift), the wrist inertia mismatch (ballasting changed nothing), torso and waist inertials disagreeing with the controller model (matching changed nothing, and the baseline's own asset disagrees too), and the apple being repelled before contact (offsets are identical; it was a real contact from a hand in the wrong place).

What still fails

Grasp slip dominates the remaining 94%: the hand reaches the apple and closes, but the contact patch is a five-finger cage against an opposed thumb rather than a three-finger pinch, and the apple rolls out during the lift. Then residual push-instead-of-lift, much reduced by the pelvis fix. Then the slower closing, which on a 6 s episode costs attempts. The first and third are what a frozen policy cannot fix.

Scaling the good episodes into training data

Because those failures are behavioural rather than geometric, the route forward is fine-tuning on target-embodiment data, and step 4 builds that without a VR rig or any teleoperation.

The idea is to keep the few episodes that already work and multiply them. Successes are harvested straight from the retargeted policy — Arena's recorder already stamps the success term and discards failures, so a rollout loop yields success-filtered demos for free. That

returned 36 successes from 553 episodes (6.5%), an independent confirmation of the measured 0.06, and those became 22 Isaac Lab Mimic source demos.

Mimic regenerates each demo against randomised scenes, which is where the scale comes from. Three randomisations are wired in: per-episode apple spawn jitter, since the stock environment spawns at a fixed pose and Mimic over a fixed scene would add only action noise; an independent seed per worker; and per-episode resampling of the grasp depth during harvesting, which raises yield and diversifies the recorded grasp, so a fine-tuned policy learns to command that grasp itself. Scaling further is mostly knobs already exposed: widen the spawn range, raise trials per source demo, feed in more harvested episodes as the success rate improves, add object and lighting variation.

Running Mimic at all meant clearing three blockers: the static task raises on the G1's dual-arm mode and ships no subtask graph, the Mimic API assumes a 23-D Pink IK action where deployment uses 50-D joint targets, and the source demos had to be rebuilt in that form. A Pink-IK embodiment wearing the Revo2 hands resolves the second by expanding the hand state through the same retargeting layer used at deployment, so the grasp in the generated data is the grasp the deployed policy makes.

Conclusion

Retargeting alone does not get this policy back to baseline. I tried the full range of strategies a frozen policy allows: closure-fraction correspondence, aperture calibration against measured fingertip openings, a searched thumb-opposition schedule, an invertible map so the policy's own command survives the round trip, friction binding, effort matching, a wrist-adapter correction to the grasp centre, and a physics audit that turned up the missing pelvis collider. Each was measured on 100 episodes where it mattered. Together they took the task from 0.00 to 0.06, and the ablation shows that no hand-side change on its own produced a single success.

That is a result rather than a failure. It says the residual gap is not geometric, because the geometry has now been matched about as well as an analytic map can match it. What is left is behavioural: a five-finger cage against an opposed thumb holds an apple differently from a three-finger pinch, and a hand that takes 0.65 s to close instead of 0.2 s arrives late to a grasp whose timing the policy learned on other hardware. Neither is addressable by further retargeting.

So the plan the work converges on is to take the few episodes that already succeed as source data, scale them with domain randomisation through Mimic, and use the result for small-scale LoRA fine-tuning rather than full post-training. That is a deliberate choice, not a shortcut: the 29 body joints are already one-to-one, the observation and action interfaces are unchanged, and the policy demonstrably knows how to do this task — what it lacks is the grasp timing and contact strategy for a different hand. A low-rank adaptation on a few hundred target-embodiment episodes is well matched to a correction of that size, and every remaining failure mode is one that demonstrations show and geometry cannot.

I did not finish the GR00T N1.7 fine-tuning run within the time available. The pipeline is built and smoke-tested end to end, the source demos are prepared and validated, and the conversion produces a dataset GR00T's own loader accepts — but generation was not run at scale and no fine-tuned checkpoint exists, so there is no post-fine-tuning number here. Everything needed to produce one is in `4_mimic_datagen/`, and the comparison to run is the same 100 episodes through `3_g1_brainco_inference/run_eval.sh` against the 0.06 measured in this report.